Maximal Subgroups
Finitely Generated Groups and Zorn's Lemma

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Overview

Theorem

For any finitely generated group, there exists a maximal proper subgroup.

1. Proof
   - Definitions: maximal element, poset, chain
   - Exercise §2.4: 17 (a,b,c)
   - Proofs
2. Discussion of Zorn's Lemma

Definitions ([D&F, 2003] Appendix I)

Definition (poset)

A *partial order* on a nonempty set A is a relation $\leq$ on A satisfying

1. $x \leq x$ for all $x \in A$ (reflexive),
2. if $x \leq y$ and $y \leq z$ then $x = y$ for all $x, y \in A$ (antisymmetric),
3. if $x \leq y$ and $y \leq z$ then $x \leq z$ for all $x, y, z \in A$ (transitive).

Definition

Let the nonempty set A be partially ordered by $\leq$.

1. A subset $B \subset A$ is called a *chain* if for all $x, y \in B$, either $x \leq y$ or $y \leq x$.
2. An *upper bound* for a subset $B \subset A$ is an element $u \in A$ such that $b \leq u$, for all $b \in B$.
3. A *maximal element* of A is an element $m \in A$ such that if $m \leq x$ for any $x \in A$, then $m = x$.

Colloquialisms ([D&F, 2003] Appendix I)

- We usually say that A is a partially ordered set under the ordering $\leq$ or that A is partially ordered by $\leq$.
- In the literature a chain is also called a *tower* or called a *totally ordered* or *linearly ordered* or *simply ordered* subset.

Exercise D&S §2.4.17: Setup

- This is an exercise involving Zorn's Lemma (see Appendix I[1]) to prove that every nontrivial finitely generated group possesses maximal subgroups.
- Let G be a group, say $G = \langle g_1, g_2, \ldots, g_n \rangle$.
- Let $\mathcal{S}$ be the set of all proper subgroups of G. Then $\mathcal{S}$ is partially ordered by inclusion.
- Let $\mathcal{C}$ be a chain in $\mathcal{S}$.

[1] We will assume Zorn's Lemma in our proof, and discuss it afterward.

Exercise [D&F, 2003] §2.4: 17: Statements

Exercise 17 (a)

Prove that the union, H, of all the subgroups in $\mathcal{C}$ is a subgroup of G.

Exercise 17 (b)

Prove that H is a *proper* subgroup. [If not, each g_i must lie in H and so must lie in some element of the chain $\mathcal{C}$. Use the definition of a chain to arrive at a contradiction.]

Exercise 17 (c)

Use Zorn's Lemma to show that $\mathcal{S}$ has a maximal element (which is, by definition, a maximal subgroup.)

Exercise [D&F, 2003] §2.4: 17: Proofs

Proof Setup (Hernandez) . . .

Let $\mathcal{C}$ be a chain in the set $\mathcal{S}$ of all proper subgroups partially ordered by inclusion of a group $G = \langle g_i \rangle_{i=1}^{n}$ generated by a finite number $n \in \mathbb{N}$ of generators.
Consider two elements of the union of all the subgroups in $\mathcal{C}$:

$$x, y \in H = \bigcup_{H_\alpha \in \mathcal{C}} H_\alpha. \tag{1}$$

Then, there exist subgroups $H_1, H_2 \in \mathcal{C}$ such that $x \in H_1$ and $y \in H_2$. By the definition of a chain, we assume without loss of generality that $H_1 \leq H_2$.

(a) Then, H is a group because it contains the following: $xy, e, x^{-1} \in \langle x, y \rangle \subset H_2 \subset H$.

Exercise [D&F, 2003] §2.4: 17: Proofs

Proof (Hernandez).

(b) Suppose that H is not a proper subgroup, namely that $H = G$. Since $g_i \in H$, there exist subgroups $H_i \in \mathcal{C}$ such that $g_i \in H_i$. By the definition of chain, there exists a permutation $\sigma \in S_n$ such that $H_{\sigma(i)} \leq H_{\sigma(j)}$ for $1 \leq i \leq j \leq n$. Then, each $g_i \in H_{\sigma(n)}$, so any $g \in G$ is in $H_{\sigma(n)}$, which contradicts its propriety. Therefore, H is a proper subgroup.

(c) According to the previous parts, $H_\alpha \in H$ for all $H_\alpha \in \mathcal{C}$. By definition, H is an upper bound for $\mathcal{C} \subset \mathcal{S}$. By Zorn's Lemma (stated below), $\mathcal{S}$ has a maximal element $M \in \mathcal{S}$. Therefore, G has a proper maximal subgroup $M \leq G$.

Lemma (Zorn's Lemma)

If A is a nonempty partially ordered set in which every chain has an upper bound then A has a maximal element.

Discussion of Zorn's Lemma ([D&F, 2003] Appendix I)

Axiom (Axiom of Choice)

The Cartesian product of any nonempty collection of nonempty sets is nonempty. In other words, if I is any nonempty (indexing) set and A_i is a nonempty set for all $i \in I$, then there exists a choice function from I to $\cup_{i \in I} A_i$, i.e. a function:

$$f : I \to \bigcup_{i \in I} A_i,$$

such that $f(i) \in A_i$ for all $i \in I$.

Principle (The Well Ordering Principle)

Every nonempty set A has a well ordering, i.e. a total ordering on A such that every nonempty subset of A has a minimum (or smallest) element, i.e., for each nonempty $B \subset A$ there is some $s \in B$ such that $s \leq b$, for all $b \in B$.

Discussion of Zorn's Lemma ([D&F, 2003] Appendix I)

Joke (Bona [Krantz, 2002])

The Axiom of Choice is obviously true, the well-ordering principle obviously false, and who can tell about Zorn's lemma?

Theorem

Assuming the usual (Zermelo-Fraenkel) axioms of set theory, the following are equivalent:

1. Zorn's Lemma
2. the Axiom of Choice
3. the Well Ordering Principle

Theorem

Zorn's Lemma is independent of the usual (Zermelo-Fraenkel) axioms of set theory in the sense that including it (xor its negation) would preserve the consistency of the axioms of set theory (if ZF is consistent).

References



David S. Dummit and Richard M. Foote

Abstract Algebra, Third Edition, Appendix I and Section 2.4

Wiley (2003).



Steven G. Krantz

The Axiom of Choice, pp. 121–126

Handbook of Logic and Proof Techniques for Computer Science, Springer (2002).